

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security

COURSE CODE: CS A5117

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

I M. Varshith Reddy(192524174) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Varshith Reddy

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 5	/ 6	/ 4	/ 3	/ 2	
2	/ 5	/ 6	/ 4	/ 3	/ 2	
3	/ 5	/ 6	/ 4	/ 3	/ 2	
4	/ 5	/ 6	/ 4	/ 3	/ 2	
5	/ 5	/ 6	/ 4	/ 3	/ 2	
OVERALL RUBRICS VALUE						
	/ 5	/ 5	/ 4	/ 4	/ 2	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
---------------	----	---	--	--	----------------------------------

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CS A5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 100

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I M. Varshith Reddy(192524174) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. Varshith Reddy

<i>Q. No</i>	<i>Understanding of Concepts (25)</i>	<i>Experimental Design (30)</i>	<i>Use of Tools (20)</i>	<i>Analysis of Results (15)</i>	<i>Documentation (10)</i>	<i>Total Score (out of 100)</i>
<i>1</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>2</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>3</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>4</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>5</i>	<i>/ 5</i>	<i>/ 6</i>	<i>/ 4</i>	<i>/ 3</i>	<i>/ 2</i>	
<i>OVERALL RUBRICS VALUE</i>						
	<i>/ 5</i>	<i>/ 5</i>	<i>/ 4</i>	<i>/ 4</i>	<i>/ 2</i>	
	<i>/ 25</i>	<i>/ 30</i>	<i>/ 20</i>	<i>/ 15</i>	<i>/ 10</i>	<i>/ 100</i>

Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results

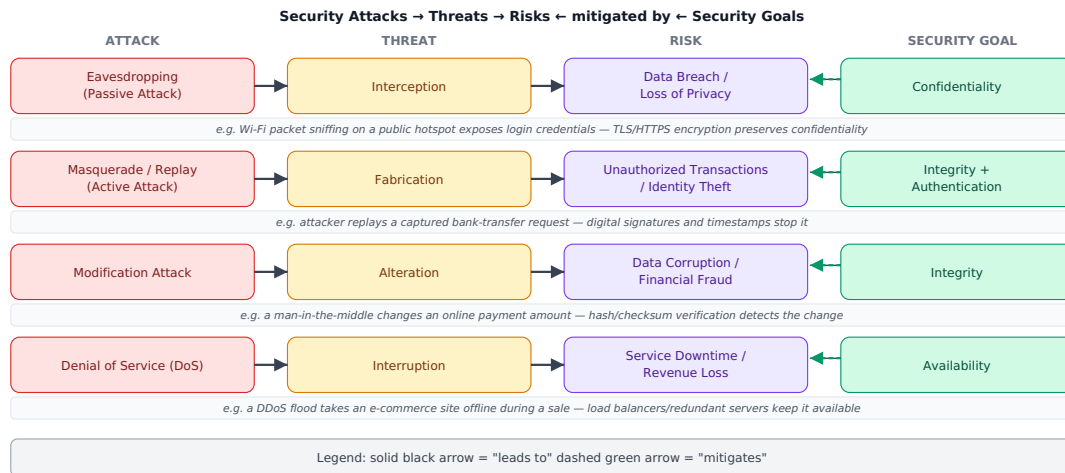
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation
---------------	----	---	--	--	----------------------------------

Cryptography and Network Security (CS A5113 / CS A5117) — Assessment Tool 2

Annexure A — Concept Mapping — CO1 Answers

Q1. Concept Map: Security Attacks, Threats, Risks and Security Goals (CIA)

Show how each attack leads to specific threats and risks; map how security goals help mitigate them; provide one real-world example for each link.



(a) How each attack leads to specific threats and risks

- **Eavesdropping (passive attack)** lets an attacker read data in transit → this realizes the **threat of interception** → the resulting **risk** is a data breach or loss of privacy (e.g., sniffing unencrypted Wi-Fi traffic to capture login credentials).
- **Masquerade / replay (active attack)** lets an attacker impersonate a legitimate party or resend a captured message → this realizes the **threat of fabrication** → the resulting **risk** is unauthorized transactions or identity theft (e.g., replaying a captured bank transfer request).
- **Modification attack** lets an attacker intercept and alter data before delivery → this realizes the **threat of alteration** → the resulting **risk** is data corruption or financial fraud (e.g., a man-in-the-middle changing a payment amount).
- **Denial of Service (DoS)** lets an attacker flood or disable a system → this realizes the **threat of interruption** → the resulting **risk** is service downtime and revenue loss (e.g., a DDoS attack taking an online store offline during peak sales).

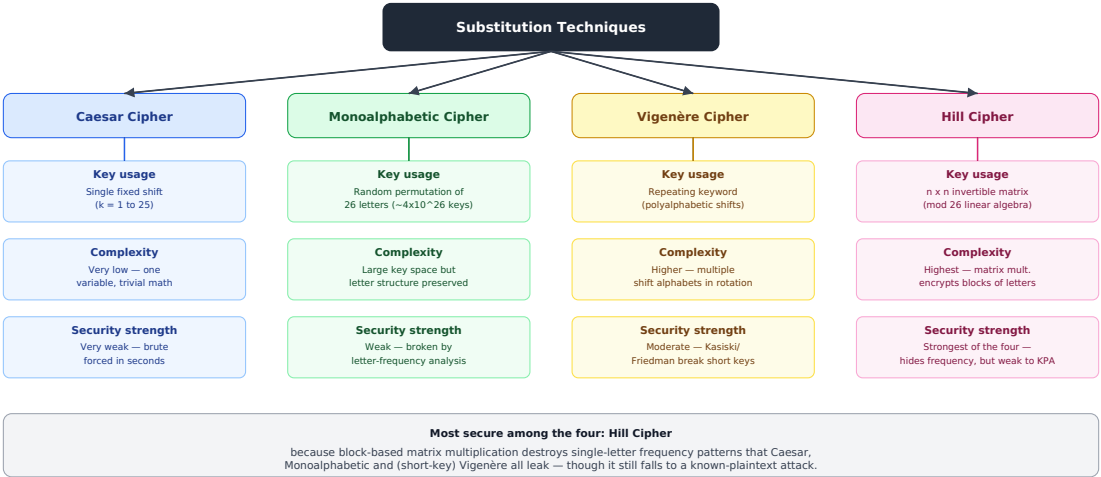
(b) How security goals mitigate the risks

- **Confidentiality** (encryption, access control) mitigates interception-based risks by making intercepted data unreadable — e.g., TLS/HTTPS protects credentials over public Wi-Fi.
- **Integrity** (hashing, digital signatures, sequence numbers/timestamps) mitigates fabrication and modification risks by letting the receiver detect forged or altered data — e.g., a bank verifies a digital signature to reject a replayed transaction.
- **Availability** (redundancy, load balancing, rate limiting, intrusion prevention) mitigates interruption risks by keeping the service running under attack — e.g., a CDN/load balancer absorbs a DDoS flood.

Summary: Each attack is a real-world action, the threat is the abstract security violation it represents, the risk is the business/operational impact, and the CIA goal is the countermeasure objective that closes the loop — this chain (Attack → Threat → Risk ← Goal) is the core of any risk-assessment concept map.

Q2. Concept Map: Comparing Substitution Techniques

Connect Caesar, Monoalphabetic, Vigenère and Hill ciphers based on Key usage, Complexity and Security strength; analyze which is most secure and why.

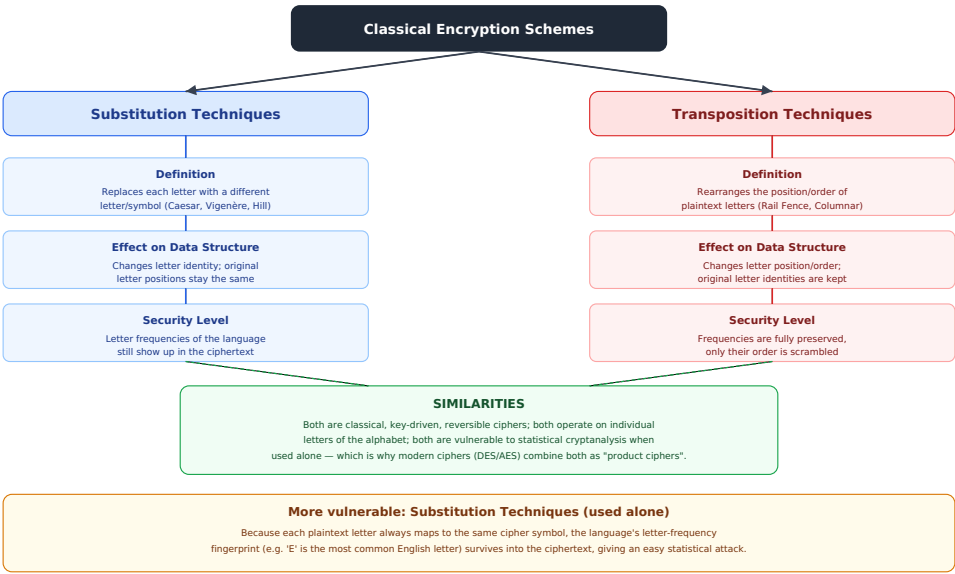


Cipher	Key Usage	Complexity	Security Strength
Caesar Cipher	Single numeric shift (1 key, 25 possibilities)	Trivial — one arithmetic operation	Very weak — exhaustive key search in seconds
Monoalphabetic Cipher	Arbitrary permutation of the 26-letter alphabet	Large key space (26! ≈ 4×10 ²⁶) but structure preserved	Weak — broken quickly using letter-frequency statistics
Vigenère Cipher	Repeating keyword drives multiple Caesar shifts	Moderate — polyalphabetic substitution	Moderate — Kasiski examination / Friedman test recover short keys; approaches one-time-pad strength only as key length → plaintext length
Hill Cipher	n×n invertible key matrix under mod 26 arithmetic	Highest — linear-algebra based block cipher	Strongest classically — resists frequency analysis, but linear structure is broken by a known-plaintext attack

Analysis — most secure technique: The **Hill Cipher** is the most secure of the four because it is a *polygraphic* substitution — it encrypts blocks of letters simultaneously through matrix multiplication, which diffuses plaintext statistics across the ciphertext and defeats simple frequency analysis (unlike Caesar and Monoalphabetic, which map one plaintext letter to one ciphertext symbol and therefore preserve language frequency patterns). The Vigenère cipher improves on simple substitution by using multiple shifting alphabets, but remains breakable once the keyword length is found. Hill's weakness is that it is fully linear, so if an attacker obtains enough known plaintext-ciphertext pairs, the key matrix can be recovered by solving simultaneous equations — a limitation none of these classical ciphers can fully avoid.

Q3. Concept Map: Substitution vs Transposition Techniques

Show differences and similarities; map how each affects data structure and security level; analyze which is more vulnerable to attacks and why.



(a) Differences and similarities

Aspect	Substitution	Transposition
Basic operation	Replaces plaintext letters with other letters/symbols	Rearranges (permutes) the positions of plaintext letters
Effect on data structure	Letter identity changes; position unchanged	Letter position changes; identity unchanged
Security level	Hides letter identity but frequency statistics leak	Hides letter order but frequency statistics remain fully intact
Similarity	Both are key-based, reversible classical techniques; both are weak alone but strong when combined (as in modern product ciphers)	

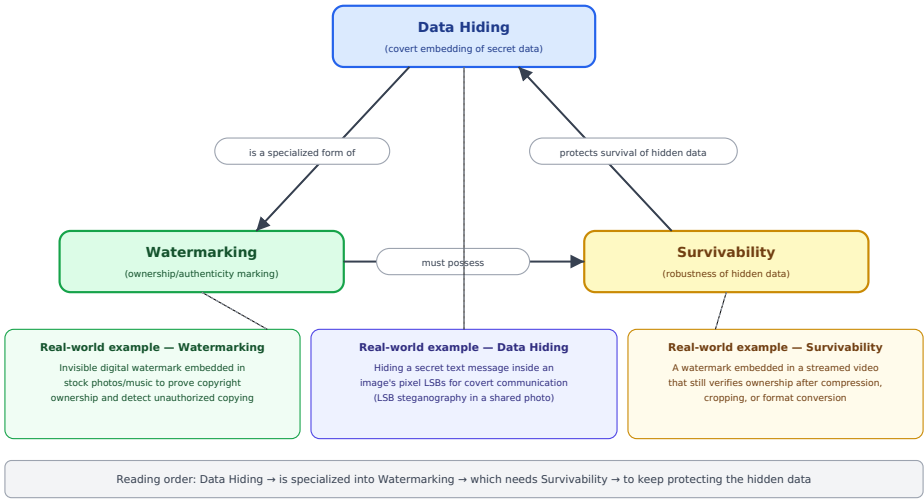
(b)(i) Effect on data structure and security level — see diagram above

(b)(ii) Which is more vulnerable, and why

Substitution techniques used alone are generally more vulnerable because a fixed letter-to-symbol mapping preserves the underlying language's frequency distribution (e.g., 'E', 'T', 'A' remain the most frequent symbols in English ciphertext), so a simple frequency-analysis attack quickly recovers the key/mapping. Transposition ciphers hide this weakness somewhat because the letter frequencies are perfectly preserved yet scrambled in position, which resists naive frequency counting but is still breakable through anagramming and pattern/column-length analysis. In practice neither is safe alone; real security comes from combining both (as later block ciphers do).

Q4. Concept Map: Steganography — Data Hiding, Watermarking, Survivability

Show relationships between these concepts; analyze how each contributes to secure communication; include one real-world application for each.



(a) Relationships between the concepts

- **Data Hiding** is the umbrella concept — concealing the very existence of information inside a cover medium (image, audio, video, text).
- **Watermarking** is a specialized, purpose-driven form of data hiding: instead of hiding a secret message, it embeds ownership/authenticity information into the cover object.
- **Survivability** (robustness) is the property that a watermark (or any hidden data) must possess to remain intact after the cover object is compressed, edited, format-converted, or attacked — it is what makes watermarking practically useful, and in turn it protects the goal of data hiding by ensuring the hidden information is not lost.

(b) Contribution to secure communication

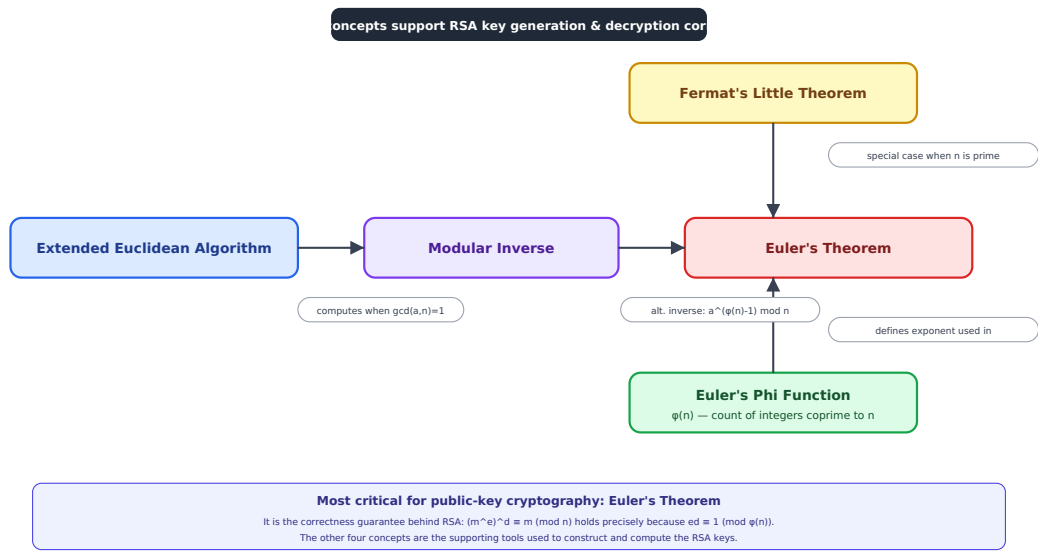
- **Data hiding** provides confidentiality of the very fact that communication is taking place (covert channel), useful when encryption alone would draw attention.
- **Watermarking** provides integrity and authenticity/non-repudiation — proving who created or owns a digital asset, deterring tampering and piracy.
- **Survivability** provides availability/reliability of the protection — the hidden data or watermark keeps doing its job even after the file is processed, compressed, or attacked, which is what makes the whole scheme trustworthy in practice.

(c) Real-world application (see diagram)

- **Data Hiding:** LSB steganography — hiding a secret message inside an ordinary-looking image for covert communication.
- **Watermarking:** Invisible watermark in stock photography/music to prove copyright ownership.
- **Survivability:** A video watermark that survives compression/cropping so ownership can still be verified after re-upload.

Q5. Concept Map: Number Theory Concepts in Cryptography

Link Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem; show mathematical connections, role in encryption/decryption, and analyze which is most critical for public-key cryptography.



(a) How the concepts are mathematically connected

- Extended Euclidean Algorithm** finds integers x, y such that $ax + ny = \gcd(a,n)$; when $\gcd(a,n) = 1$, $x \pmod n$ is the **Modular Inverse** of $a \pmod n$.
- Euler's Phi Function** $\phi(n)$ counts integers less than n that are coprime to n ; it is the exponent base used inside **Euler's Theorem**: $a^{\phi(n)} \equiv 1 \pmod n$ for $\gcd(a,n) = 1$.
- Fermat's Little Theorem** ($a^{p-1} \equiv 1 \pmod p$, for prime p) is a special case of Euler's Theorem when $n = p$ is prime, since $\phi(p) = p - 1$.
- Euler's Theorem gives a second way to compute the **Modular Inverse**: $a^{\phi(n)-1} \pmod n = a^{-1} \pmod n$, complementing the Extended Euclidean Algorithm.

(b) Role in encryption/decryption

- Euler's Phi Function** computes $\phi(n) = (p-1)(q-1)$ from the RSA primes p, q — needed to choose the key pair.
- Extended Euclidean Algorithm** computes the private key d as the modular inverse of the public exponent e , i.e., $d \equiv e^{-1} \pmod{\phi(n)}$.
- Euler's Theorem** mathematically guarantees that decryption recovers the original message: $(m^e)^d \pmod n = m$, because $ed \equiv 1 \pmod{\phi(n)}$.
- Fermat's Little Theorem** is used in primality testing (to efficiently verify that p and q are prime before they are used) and underlies fast modular exponentiation optimizations (CRT-RSA).
- Modular Inverse** itself is the concrete number (the private key d , or coefficients in CRT decryption) that these theorems/algorithms exist to compute.

(c) Which concept is most critical for public-key cryptography, and why

Euler's Theorem is the most critical concept. While Euler's Phi Function and the Extended Euclidean Algorithm are essential *tools* for constructing the RSA key pair, it is Euler's Theorem that supplies the actual mathematical proof that encryption followed by decryption returns the original plaintext. Without this theorem, there would be no guarantee that choosing $d = e^{-1} \pmod{\phi(n)}$ actually inverts the encryption operation — it is the correctness foundation on which the entire RSA public-key scheme rests. Fermat's Little Theorem matters practically (primality testing, faster computation) but is only a special case of the more general Euler's Theorem.